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Reducing Patient Waiting Time at Public Hospitals through Predictive Modeling

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***Abstract*— Reducing patient waiting times in public hospitals is a critical challenge that significantly impacts patient satisfaction, healthcare outcomes, and operational efficiency. Efficient, effective, and timely patient care contributes toward patient service satisfaction. This is not always true for most healthcare institutions. This paper investigates the use of predictive modeling in addressing patient waiting time by leveraging historical data from electronic records at five selected public health centers in Harare, Bulawayo, and Gweru between October 1 and 31 December 2023. The objective was to identify a predictive model that could effectively classify the queue data into two categories: delayed (Yes) or not delayed (No). Logistic Regression, Decision Tree, Random Forest, and XGBoost models were trained using Stratified K-Fold cross-validation to determine the model with the best fit on the original data. The models were trained using variables such as queue position, wait time, service time, location of the health center, and weekday as predictors. XGBoost emerged as the top model, achieving the highest ROC AUC of 0.63, compared to the other models which ranged from 0.56 to 0.6. The selected model was then evaluated using F1 Score, ROC Curve, and Precision-Recall Curve. Results showed a high F1 Score of 0.702, indicating the model's ability to distinguish between the positive (Delayed) and negative (Not Delayed) classes. The area under the ROC curve (AUC) of 0.63 further quantified the model's overall ability to discriminate between these classes. Additionally, the area under the Precision-Recall curve (AUC-PR) of 0.64 suggests that the model performs well in balancing precision and recall, effectively identifying most actual delayed services (high recall). Accurately predicting patient waiting times not only enhances patient care experience but also supports healthcare providers in delivering timely and efficient care. Designers of hospital programs can analyze patterns, and identify peak times, predict high-delay patient arrivals, and suggest real-time staffing adjustments. Future work will focus on refining the models and integrating them into hospital management systems for broader implementation.**

Keywords— Patient waiting time, XGBoost, public health institution, quality health care

I. Introduction

Patient waiting times in hospitals, particularly in outpatient departments and life-threatening care services, have received ongoing focus for decades. Effective and appropriate care for patient satisfaction is guaranteed if patients are content. Patient contentment is a prerequisite for professionals to stay competitive in the healthcare market [1]–[4]. Given this, organizations must embrace new technology that supports the highest caliber of contemporary healthcare [5]. Artificial Intelligence seems to be the promising technology of the century regarding the modeling of queues in general.

The use of machine learning-based healthcare solutions in Africa is still growing [6][7]. Although systematic and standardized health systems are fast increasing in several Southern African nations, including Zimbabwe, as described by [4], Southern Africa is still to fully benefit from investing in machine learning models in health care. The healthcare system, if well supported by digital technology, is likely to improve in terms of its capacity, effectiveness, and quality. Technology presents an opportunity for health professionals to promote health and improve health equity among patients. According to the World Health Organization (WHO) [3], [8] quality of care includes among other factors, reduction of patient waiting times in the Outpatient Department (OPD). This can be achieved by coordinating, and communicating with all stakeholders in an efficient and timely manner [6]. Further recommendations for quality health care include effective, safe, and people-centered services [4], [9]. The 31 informed plans by easing the difficulties associated with locating, comprehending, utilizing, and evaluating health information and services [10].

Public health institutions in Zimbabwe, provide healthcare to the general public which includes individuals, families, and companies. Most public hospitals aim to provide quality health care at affordable rates of charges. To ensure proper estimates of patient waiting times and minimal scheduled appointment delays as a means to enhance patient happiness and enable personnel to monitor and respond to patient flow more precisely, public hospitals need to invest in meaningful solutions to the problem of elongated patient waiting times. The suggested innovation in this paper is a machine learning model that public hospitals can apply to their patient management processes, especially to help predict patient waiting times and enable the enhancement of standard business processes. Machine learning is one of the most practical and efficient ways to deal with the extreme complexity and unpredictable nature of waiting and delay time patterns [1], [10]. What standard regression analysis cannot find in complicated processes, machine learning models can [11]. The latter can adjust to changing conditions, tolerate noise and unusual outliers, and can function without human supervision. Consequently, the ultimate objective of this effort was to create a machine-learning model that improves public hospitals' forecast accuracy for both scheduled and walk-in patient visits. The model provides a forecast for patient waiting times enabling the healthcare stakeholders to plan effectively. The code for model implementation is available on https://github.com/marydzinomwa/Reducing-patient-waiting-times-in-public-hospitals-using-predictive-modelling

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In the subsequent sections, we discuss the development of the waiting time prediction model. In Section II we review the literature related to quality of care and waiting time as a determinant of quality of care. In the same section, we review some innovations designed to tackle patient waiting times, with machine learning applications constituting the bulk of the review. In Section III we discuss the methodology used to build the model, followed by the presentation of an in-depth discussion of the results in Section IV. Section V concludes the study, and presents recommendations for further work

II. Patient Wait Times and Quality Health Care

In designing models to tackle the problem of patient waiting times, it is important to uncover the factors affecting patient wait times, as well as the interplay among these factors. It is also important to identify solutions that have been adopted to optimize patient waiting times. Machine learning models have been chosen in this paper since the literature shows that they have been applied successfully in this area [1], [10]–[13]. From this review, we select the most appropriate and applicable machine learning pipeline, machine learning algorithms, model evaluation techniques and then propose a machine learning model that can predict and optimise waiting time in outpatient departments for public hospitals.

A. Factors Affecting Patient Waiting Times

Patients and healthcare providers in Africa continue to face difficulties due to lengthy waiting times in outpatient departments [2]. A study by [7] concluded that patient long waiting times were one of the causes of low-quality patient care in a public hospital in Southern Ethiopia. Prolonged waiting periods can result in missed appointments and treatment delay, and in extreme cases death of patients in queues. Waiting times influence patients' opinions of the clinic's experience and service delivery [3], [4]. Several factors are key in influencing patients' waiting times. Each of the factors needs appropriate strategies to lower the waiting time. Among the factors influencing patient waiting time we have: appointment scheduling, referral pathway and administrative processes, triage process, technological infrastructure, frequency of emergency cases, patient complexity, resource allocation, communication and information flow, and presence of quality improvement initiatives [1], [10], [11], [14], [15]. We discuss some of these in the subsequent paragraphs.

Regarding appointment scheduling, the platform used for making the reservation plays a big role on the time the patient will wait to get a service. Patients making their appointments through electronic means tend to be time-conscious, arriving at the hospital on time, reducing the time they wait in the queue [8]. The number of patients scheduled per health centre also plays a role in determining how long a patient will wait to get their attention. This is also further impacted by the number of health service providers in the centre [16], [17]. In developing countries, the human resource base is often limited just as the health equipment necessary for quality patient care.

Another determinant in patient waiting time is the referral pathways. Streamlined administrative processes play a crucial role in improving patient waiting times within the healthcare system [14]. Clearly defined referral pathways help healthcare providers establish a more organized and optimized patient flow. When there is a well-structured system for referrals from primary care to specialists or between different healthcare facilities, patients can receive timely and appropriate care without unnecessary delays. It is critical to identify and address problems that impact patients' access to specialty care for non-emergent procedures and how these inadequacies can be addressed through interventions in the delivery of health to improve care for patients with chronic conditions. Value Stream Mapping (VSM), a lean technique, was used by [14], who investigated referral pathway processes and recommended digital solutions to reduce delays in patient journeys in hospitals [15].

On a different note, Newman-Casey [18], notes that most outpatient departments in clinics do not have the infrastructure required to precisely measure and record patient waiting times. In tackling this problem, the authors recommend the construction of inexpensive, portable, passive real-time locating devices for their outpatient departments to monitor patient wait times and service provider-patient interactions. Other recommendations include expanding to peak, patient service-provision based on demand [18]. However, of these services, many stay unused, creating an unsustainable cost structure. As a result, a more useful strategy has been to apply queuing theory and other operations management strategies to identify the ideal location were minimizing both service costs and client waiting times is possible. The use of self-service technology is a crucial strategy for cutting down on patient waiting times in the delivery of contemporary healthcare services as shown by [9].

In the following paragraphs we discuss measures to address the problem of long waiting times in general, and in public hospitals in particular.

B. Machine Learning in Tackling Patient Waiting Times

Patient waiting time in health care has forced researchers to turn to artificial intelligence for solutions. Some of the trusted technologies include machine learning. With regards to machine learning, it is important to consider machine learning pipelines, feature engineering, pre-processing, and data validation [18]. A machine learning pipeline, is an end-to-end framework that synchronizes the flow of data into a machine learning model [10]. This usually involves several sequential processes, from data extraction and preprocessing to model training and deployment, as seen in Fig 1.

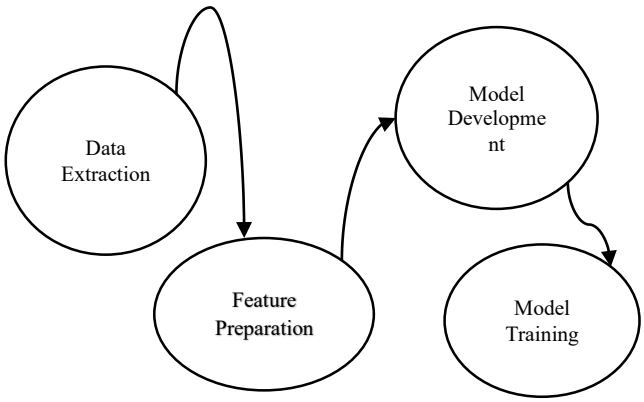


Fig 1.Machine Learning Pipeline1

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Data pre-processing can either be manual or automated. An automated cycle continuously carries out model training by automating the Machine Learning (ML) pipeline. The cycle automates the procedure that generates new data for retraining the model in production. While AutoML systems may produce predictive models quickly and reach near-optimal results [19], their coverage is still limited and their true potential remains untapped, therefore this study will focus on the manual cycle.

There is no true infrastructure to support the automation of data collection, data cleansing, model training, and evaluation in a manual workflow [18]. The machine learning pipeline workflow is a script-based, manual process that includes steps including data preparation, analysis, training, testing, and deployment. This way the process does not need a DevOps (additional Developers and Operations) team because there is no automatic feedback loop and switching between procedures is done manually [20].

C. Machine Learning Algorithms

Machine learning algorithms exist in three flavors: supervised, semi-supervised, and unsupervised learning [21]. In a supervised machine learning method, all data are labeled. The algorithm learns through building a function that maps new data on an expected output value. In an unsupervised learning algorithm, all data are unlabeled, and the system learns from the input data's inherent structure. A semi-supervised machine learning problem falls somewhere in the middle between supervised and unsupervised learning. Our interest in this paper is on supervised machine learning.

1) Supervised Machine Learning Algorithms

A supervised machine learning algorithm uses a collection of training data with labels supplied to it to develop a rule that can accurately predict the labels for new observations [18]. The algorithm is tasked with iteratively making predictions on the training data and, getting corrections during the process when erroneous guesses are made. The correct answers are known during learning. Learning stops when the algorithm performs at a level that is deemed acceptable. Supervised machine learning models are used to address either regression or classification issues, hence they are divided into those two categories. A discrete class label is predicted in a classification task, but a continuous quantity is predicted in a regression problem. However, both approaches share the same principle of making predictions using known variables.

Training classification models have made use of a number of supervised machine-learning methods. Examples of frequently used supervised machine learning techniques that address classification problems are: Linear Classifiers, Neural Networks, Decision Trees, Support Vector Machines, and Nave Bayes Classifiers [22]. Supervised machine learning algorithms, which learn from labeled training data, can be applied to optimize patient waiting times in healthcare settings by predicting and understanding factors influencing delays

2) Linear Classifiers

A Linear Classifier method classifies input vectors using linear (hyperplane) decision bounds. Items with similar feature values are attempted to be grouped using a linear classifier approach. According to Wang (2021), a linear classifier makes classifications based on the value of the linear combination of the attributes in the dataset. Using a linear classifier has the benefit of processing data rapidly. Several studies have shown that this method is among the quickest classifiers. Additionally, linear classifiers generally perform better when the number of dimensions is quite large, such as is the case with document classification, where each element is frequently the number of times a word appears in a text. On the other hand, the margin determines the rate of convergence among the variables in the data set. Generally speaking, the margin measures a dataset's linear seperability and, consequently, the ease with which a particular classification problem may be resolved using the approach. Linear classifiers can analyze historical data and patient characteristics to predict the likelihood of a patient arriving on time for appointments. By categorizing patients into groups based on the predicted probability of punctuality, healthcare providers can optimize appointment scheduling, allocate resources efficiently, and reduce wait times

3) Naïve Bayes Classifiers

The Naïve Bayes method is a supervised learning approach that addresses classification problems by applying the Bayes theorem [23]. It is among the best classification algorithms for quickly creating machine learning models that can predict the future. Several studies have found the Naïve Bayes Classifier to be one of the most basic, yet consistently effective, classifier models. However, the Naïve Bayes Classifier's presumption that every attribute used to characterize an instance is conditionally independent given the instance's class poses an inherent issue when "information is double counted" and interaction omission occurs, which can lower classification accuracy. In terms of patient scheduling, the Naive Bayes classifier can be employed to predict the likelihood of a patient arriving for an appointment based on historical data and various features (e.g., appointment time, day of the week, patient demographics). By using this prediction, healthcare providers can optimize appointment scheduling, allocate resources more efficiently, and minimize no-shows, which can contribute to reduced wait time.

Regarding resource allocation, the Naive Bayes classifier can be used to predict the demand for healthcare resources, such as the number of patients expected at a particular time. By anticipating resource needs, healthcare facilities can adjust staffing levels, optimize workflow, and allocate resources effectively, leading to a more streamlined patient flow and potentially reduced wait times.

4) Support Vector Machines (SVM)

Support Vector Machines (SVMs), use classification methods to address two-group classification issues. Finding the perfect hyperplane in the training data space that represents every observation in the dataset [12] is the main objective of support vector machines (SVMs). The line used to anticipate the target is called a hyperplane. It is possible for the support vectors or data points closest to the boundary lines to be both within and outside of them. Next, any equation may be used to establish the hyperplane, whether it be non-linear or polynomial. SVM is useful for classification and regression tasks. In the context of patient wait times, and like the Naive Bayes Classifier, SVM can predict whether a patient is likely to experience extended wait times based on features like appointment type, historical data, or time-related variables. This information can inform resource allocation strategies. Compared to other supervised machine learning techniques, such as neural networks, SVMs have two main advantages: they are faster and perform better with less amounts of data.

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5) Decision Trees

A Decision Tree is a supervised machine-learning algorithm that represents all possible solutions in a graphical framework [26]. All decisions are made based on predetermined criteria. [27] created a Chatbot or virtual assistant that used a dataset of previously captured basic questions and answers, or FAQs, to diagnose ailments based on user queries. To train the model, a decision tree method was utilized, which relied on a very diverse and accurate database of symptoms and disorders. When employing the chatbot, a user asks a query and receives a preferable and highly correct diagnosis.

Decision tree algorithms, including Random Forests can be employed to understand the hierarchy of factors influencing patient wait times. Decision trees provide insights into critical decision points and help healthcare providers prioritize interventions for the most impactful improvements.

6) Random Forests

Random Forest is one of the most popular and widely used algorithms in classification and regression applications. Using several data, it builds decision trees, using the average for regression and the majority vote for categorization. In contrast to a Decision Tree, which creates rules and predictions using a feature-rich data set as input, a Random Forest gathers observations at random, constructs a decision tree, and then computes the average output without the need for any formulas. [28] used a random Forest algorithm to perform an intent classification problem for high school students' inquiries. The data focused on high school kids' anxieties about their future; hence, the chatbot supported them with rapid support to prepare them to make the proper profession or career choices.

Random Forest (RF) is a powerful ensemble learning algorithm that can be applied to optimize patient waiting times in healthcare settings. RF builds multiple decision trees during training and merges their predictions to improve accuracy and robustness. Here are ways Random Forest can be utilized to optimize patient waiting times. RF can help identify the most influential factors contributing to patient wait times. By understanding these predictors, healthcare providers can implement targeted quality improvement initiatives, process optimizations, or interventions to address specific issues and reduce delays.

7) K-Nearest Neighbor (K-NN)

K-Nearest Neighbor (KNN) is a simple and effective algorithm for classification and regression tasks, and is a powerful prediction model [23]. By predicting patient waiting times based on the characteristics of similar historical cases, KNN can offer insights into patterns and trends that contribute to delays. When implementing supervised machine learning algorithms in healthcare, it is essential to consider ethical implications, data privacy, and collaboration with healthcare professionals to ensure that models align with clinical expertise. Additionally, ongoing validation and refinement are crucial to adapt models to evolving healthcare dynamics. The integration of machine learning predictions into decision-making processes can ultimately lead to more efficient and patient-centred care with optimized wait times.

D. Machine Learning Model Evaluation Techniques

The best Machine Learning Model Evaluation technique should be chosen based on its suitability to a given dataset for solving a certain problem, and it is referred to as the "Best Fit" because it outperformed other strategies based on the same input dataset [22]. To evaluate the performance of a machine learning model, the following metrics can be used: Classification Accuracy, Confusion Matrix, Logarithmic Loss, Area under Curve (AUC), and F-Measure.

Reviewing the literature led to a closer look into relevant machine learning algorithms, and the machine learning models that were previously applied in relation to patient waiting time optimization problems. This was critical to expanding the researchers' understanding and appreciation of available alternatives. The literature review was also important in gaining beneficial insights from other scholars' algorithm selections and the performance outcomes acquired from each algorithm, technique, and model applied. Equipped with technologies and existing model implementation, in the next section we discuss the methodology used to design the proposed model for optimizing patient waiting times..

III. RESEARCH METHODOLOGY

The research philosophy guiding this study was positivism, recognizing the view that knowledge can only be acquired through empirical research, which is based on measurement and observation. In other words, this study is based on the view that all knowledge is regarded as a posteriori knowledge – knowledge that is not reliant on human reasoning but instead is gained from research. Therefore, to address the complex problem of optimizing waiting times in the outpatient department (OPD) of selected public hospitals, the research philosophy adopted was positivism, purely concluding and making recommendations based on the machine learning models performance and findings in relation to the literature on related studies.

We utilized a repeated cross-sectional (RCS) design, in which data was collected from independent participants at multiple points in time. The researchers also made use of quantitative research methods, with an analytical study design. Unlike a longitudinal study, where the same individuals are followed over time, RCS studies focus on different groups of participants at each wave. In other words, it is a process of taking snapshots of different populations at different moments.

In the context of this study, we considered the cost-effectiveness of the methodology: An RCS study is more cost-effective than a longitudinal study because it does not require following the same individuals over time. The database from which the data was extracted was not tracking patient-level data longitudinally. In addition, we considered flexibility, that is, the method allowed new subjects to be recruited for each wave, allowing the study to have Statistical Power, that is, as new data is continually collected, the sample size remains adequate for analysis. The researchers applied RCS using data on patients who came for different health services at selected public hospitals in Harare, Bulawayo, and Gweru, Zimbabwe between 1 October and 31 December 2023.

The first author collected data on patient waiting times, and factors affecting wait times (appointment scheduling, triage processes), as well as patient demographics. This was due to the accessibility of this data to them. Data was collected at different time points from the places of interest. By comparing waiting times across different time points, the methodology allowed the researchers to identify trends and potential areas for improvement. In the context of optimizing waiting time, quantitative methods allow the researcher to

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measure and quantify various factors related to patient waiting times. By using quantitative data, we can identify patterns, correlations, and potential causal relationships that inform decision-making. Snapshots of waiting times and associated factors were collected for a period of one month, that is 1 to 31 April 2024. By examining data from a single point in time, this approach provided insights into current practices and challenges within the OPD. Quantitative techniques focusing on retrospective analysis of electronic health records and appointment scheduling systems were used to analyze the data.

In compliance with ethical requirements, the researchers sought approval from the Board of Directors overseeing the hospitals involved in the study, as well as ethical clearance from the National University of Science Technology (NUST) Research Ethical Committee. This was necessary for safeguarding the confidentiality of the data of human subjects that provided the data. All data handling procedures complied with relevant regulations and ethical guidelines to ensure confidentiality and integrity.

Ethical considerations are paramount when collecting and utilizing patient data. In this regard, the researchers deliberately considered the following key aspects to ensure ethical practices. The study utilized secondary, electronic data. There was no direct interface with human subjects, and therefore no informed consent was sought. To ensure equity and fairness, the electronic dataset used did not disproportionately exclude certain groups as all patient-level data were included in the study, regardless of patient health and social status (for example medical aid classification). To achieve patient privacy and confidentiality, the researchers safeguarded patients' sensitive information by ensuring that only authorized personnel had access to the data. For anonymization and de-identification, the data extracted and used excluded personally identifiable information (PII), and that enabled the protection of patient identities. In addition, the researchers considered data security in terms of its storage. The researchers made use of participating hospitals, existing databases which are secure enough to prevent unauthorized access or breaches.

At all participating hospitals across the country, the norm is that observation data of patient arrivals, service duration, and departure are recorded for patients consulting a specialist in the OPD unit. Data collection for this study therefore involved accessing electronic records and databases maintained by the organization upon approval by the Head of Department - ICTs. Data was extracted strictly for patients who came to the OPD and from four selected service delivery sites in Bulawayo, a site in Gweru, and a site in Harare. We extracted data with the 16 variables shown in Table 1. Each variable is described in the table under feature description..

Before applying machine learning algorithms, the collected data went through pre-processing to ensure its quality and suitability for analysis. The pre-processing steps included: Identifying any missing values in the dataset, which were addressed through the removal of incomplete records. Jupiter Notebooks was used for the purpose. In carrying out feature engineering, one additional feature was derived from the existing variables to enhance the predictive power of the model. The derived feature is waiting time which is calculated as the difference between appointment times and actual arrival times.

Table 1 PATIENT DATA EXTRACTED FROM ELECTRONIC RECORDS

Feature Name	Feature Name
Region	City where the clinic or hospital is located
PatientId	Identification of a patient
Branch	The hospital or clinic to which the entry belongs
Sex	Male or Female.
Age	How old is the patient.
p_ArrivalDate	The day of the patient's visit to the health facility, when visits the doctor.
p_ArrivalTime	The time at which the patient arrived at the health facility.
p_ConsultationTime_0	The time at which the appointment begins.
Is_on_Medical_aid	True or False. Indicates whether the patient is on a medical aid society accepted by PSMI.
digestive_system_disorder	True or False. Indicates if the patient has digestive system disorder.
Infectious_and_parasitic	True or False. Indicates if the patient has infectious and/or parasitic infection
respiratory	True or False. Indicates if the patient is a respiratory challenge
genitourinary	True or False. Indicates if the patient has genitourinary challenges
other_diseases	True or False. Indicates if the patient has other diseases
lineCounnt_0	Number of patients in queue at the time of arrival
p_wait_time (Mins)	Derived: p_ConsultationTime_0 Minus p_ArrivalTime
Delayed	True or False (Target variable). Indicates if the patient was attended to promptly (0) or was delayed (1)

For the purposes of data normalization, numeric variables were scaled to a common range to prevent biases in model training due to differences in magnitude. A MLP classifier, which is a kind of artificial neural network, was used to optimize patient waiting times in this study.

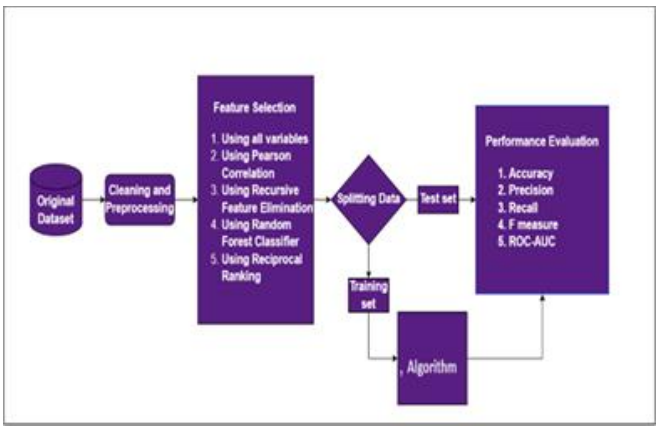


Fig 2. Applying the MLP Method

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Table 2 Model Performance Evaluation Metrics

Performance Measure	Explanation	Mathematical Formula
Mean Absolute Error (MAE)	MAE is a measure of the average magnitude of errors between paired observations.	$(1/n) * \Sigma \lvert y^i - \hat{y}^i \rvert$ where n is the number of observations and y^i is the ith observed value and $\hat{y}^i$ the ith predicted value.
Mean Squared Error (MSE)	Used to measure model accuracy	$(1/n) * \Sigma_{i=0}(y^i - \hat{y}^i)^2$
Root Mean Squared Error (RMSE)	Measures the average difference between the predicted values and the actual values in a dataset	Sqrt $(1/n) * \Sigma_{i=0}(y^i - \hat{y}^i)^2)$ where n is the number of observations and y^i the ith observed value and $\hat{y}^i$ the ith predicted value
R – squared (R2) score	Used to evaluate model quality	$R^2 = 1 - \Sigma(yi - \hat{y}i)^2 / \Sigma(yi - \bar{y})^2$ where n is the number of observations and y^i the ith observed value and $\hat{y}^i$ the ith predicted value

Prior to training the predictive model, the dataset was split into separate sets for training and testing. The training set was used to train the model, and the testing set was used to assess the final performance of the trained model on unseen data. The splitting criteria used was 80-20 for training and testing respectively.

The trained models were evaluated using the metrics Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R2) score. Table 2 explains these metrics and the formulae associated with them. These metrics quantify the difference between the predicted wait times ($\hat{y}^i$) and the actual wait times (y^i) in the testing dataset. A lower error value or a higher R2 score indicates better performance of the model in predicting patient wait times.

The chosen model was further evaluated for Accuracy, Precision, Sensitivity, F measure, and Area under the Receiver Operating Characteristic curve. The researcher fine-tuned the hyperparameters of the selected model using Hyperopt. The objective was to maximize the ROC AUC score. Hyperopt employs Bayesian optimization to find the best hyperparameters more efficiently than grid search or random search.

By following these steps, healthcare organizations can develop accurate predictive models using MLP classifiers to optimize patient wait times, thereby improving resource allocation and enhancing overall patient satisfaction. Once the optimal machine learning model is identified, it will be implemented in the outpatient department of the participating hospitals. The implementation plan involves integrating the model into the existing scheduling system to provide real-time recommendations for optimizing wait times.

IV. Results

A. Results of Exploratory Data Analysis

Exploratory data analysis (EDA) and data visualization were employed to summarize, visualize, and thereby explore historical patients' data in preparation for dimensionality reduction. EDA provided a window to gain some insights into the patients' data, such as its distribution, outliers, correlations, and missing values. EDA also helped in the identification of potential features for the proposed machine learning model. The process utilized various python tools and libraries, including, but not limited to, pandas, matplotlib, seaborn, and plotly. Fig.3 to Fig..6 show some sample graphs used to visualize EDA. In this paper we have omitted data relating to the named hospitals due to privacy issues.

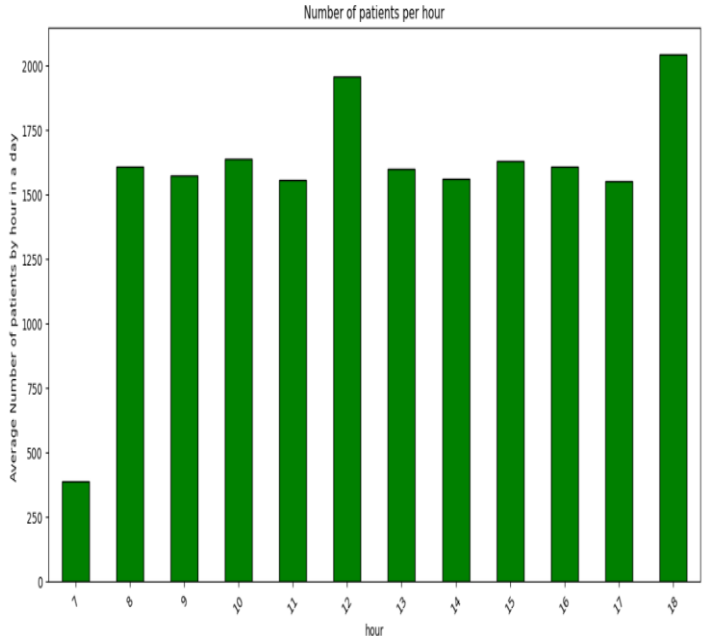


Fig 3. Number of Patients per Hour

Data visualization techniques provide intuitive ways to explore complex healthcare datasets, identify patterns, and gain insights into underlying relationships. Visualization methods such as scatter plots, histograms, and heatmaps help researchers understand the distribution, variability, and correlations among variables.

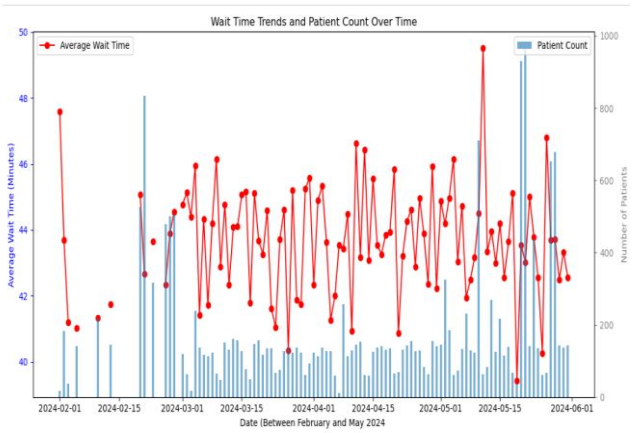


Fig 4. Trends in Patient Count over Time

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XGBoost emerged as the top model, achieving the highest ROC AUC of 0.63, compared to the other models which ranged from 0.56 to 0.6., and was the selected model in this research. The selected model was then evaluated using F1 Score, ROC Curve, and Precision-Recall Curve. Results showed a high F1 Score of 0.702, indicating the model's ability to distinguish between the positive (Delayed) and negative (Not Delayed) classes. The area under the ROC curve (AUC) of 0.63 further quantified the model's overall ability to discriminate between these classes. Additionally, the area under the Precision-Recall curve (AUC-PR) of 0.64 suggests that the model performs well in balancing precision and recall, effectively identifying most actual delayed services (high recall). Our selected model option XGBoost is therefore ideal for this study due to its ability to handle noisy data. Medical data collected from patients can often be noisy as it tends to be incomplete, and inconsistent. Further, data cleaning procedures applied may not have been fully adequate to address all noise in the data. In addition, one of the favourable features of XGBoost is its ability to handle overfitting through regularization as it incorporates both L1 (Lasso) and L2 (Ridge) regularization techniques, which help in penalizing complex models and thus prevent overfitting.

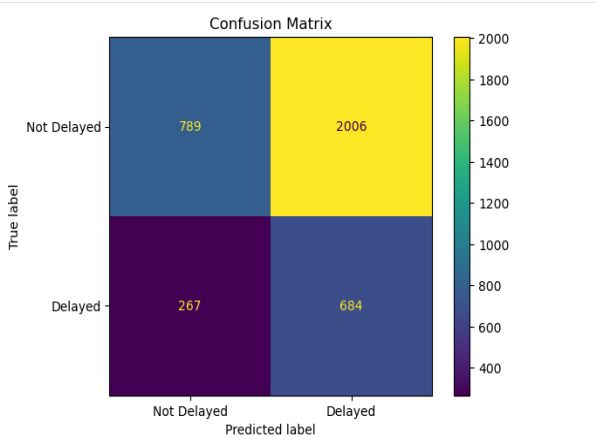


Fig 5. Confusion Matrix

The confusion matrix (Fig. 5) shows the true Negative (TN): 789 - the number of actual "Not Delayed" cases that were correctly predicted as "Not Delayed". At the same time, False Positive (FP): 2006 - the number of actual "Not Delayed" cases that were incorrectly predicted as "Delayed". False Negative (FN): 267 - the number of actual "Delayed" cases that were incorrectly predicted as "Not Delayed", and True Positive (TP): 684 - the number of actual "Delayed" cases that were correctly predicted as "Delayed".

Classification Report:

	precision	recall	f1-score	support
Delayed	0.75	0.28	0.41	2795
Not Delayed	0.25	0.72	0.38	951
accuracy			0.39	3746
macro avg	0.50	0.50	0.39	3746
weighted avg	0.62	0.39	0.40	3746

Fig 6. Model Classification Report

As shown in Fig 6, the model has high precision but lower recall for the "Delayed Service" class. For the "Not Delayed" class, the model has high recall but low precision, meaning it identifies most of the "Not Delayed" but also has a lot of false positives.

B. Discussion of Results

Waiting time can be used to determine a hospital's quality of service. This time is complex in that it is indeterminant and difficult to predict. This paper develops an ideal model for predicting waiting time at public health institutions. From the results, it is clear that the XGBoost algorithm, often referred to as "eXtreme Gradient Boosting," had the best results. XGBoost emerged as the top model, achieving the highest ROC AUC of 0.63, compared to the other models which ranged from 0.56 to 0.6. The selected model was then evaluated using F1 Score, ROC Curve, and Precision-Recall Curve. Results showed a high F1 Score of 0.702, indicating the model's ability to distinguish between the positive (Delayed) and negative (Not Delayed) classes. The area under the ROC curve (AUC) of 0.63 further quantified the model's overall ability to discriminate between these classes. Additionally, the area under the Precision-Recall curve (AUC-PR) of 0.64 suggests that the model performs well in balancing precision and recall, effectively identifying most actual delayed services (high recall). Accurately predicting patient waiting times not only enhances patient care experience but also supports healthcare providers in delivering timely and efficient care. This is in agreement with the literature as the algorithm is mainly used for classification problems due to several key features like regularization. Its favorable features include techniques to prevent overfitting, which is crucial for maintaining the model's generalization. Another feature like cross-validation: ensures that the model performs well on unpredictable data. Further, XGBoost can automatically handle missing data, which is a common issue in real-world datasets.

Also, XGBoost supports various objective functions and evaluation criteria, making it adaptable to different kinds of classification problems. These features and other unnamed features contribute to XGBoost's popularity and effectiveness in solving various classification challenges in machine learning, making it an ideal algorithm to use in the healthcare industry.

V. Conclusion

From the analysis of patients visiting the OPDs in the three-month period spanning February to May 2024, it was observed that delays tend to be high during the last days of the month. Subsequent investigation revealed the correlation among patient arrival time, patient consultation time, and the possibility of service delay. The OPDs can therefore optimize waiting times by sending SMS or WhatsApp reminders to patients with appointments. Although the average patient waiting time was not correlated to the day of week, it was observed that more patients visit the OPDs during weekdays, especially Mondays and Tuesdays. This would be a hint to the management to ensure adequate resource availability during these days.

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